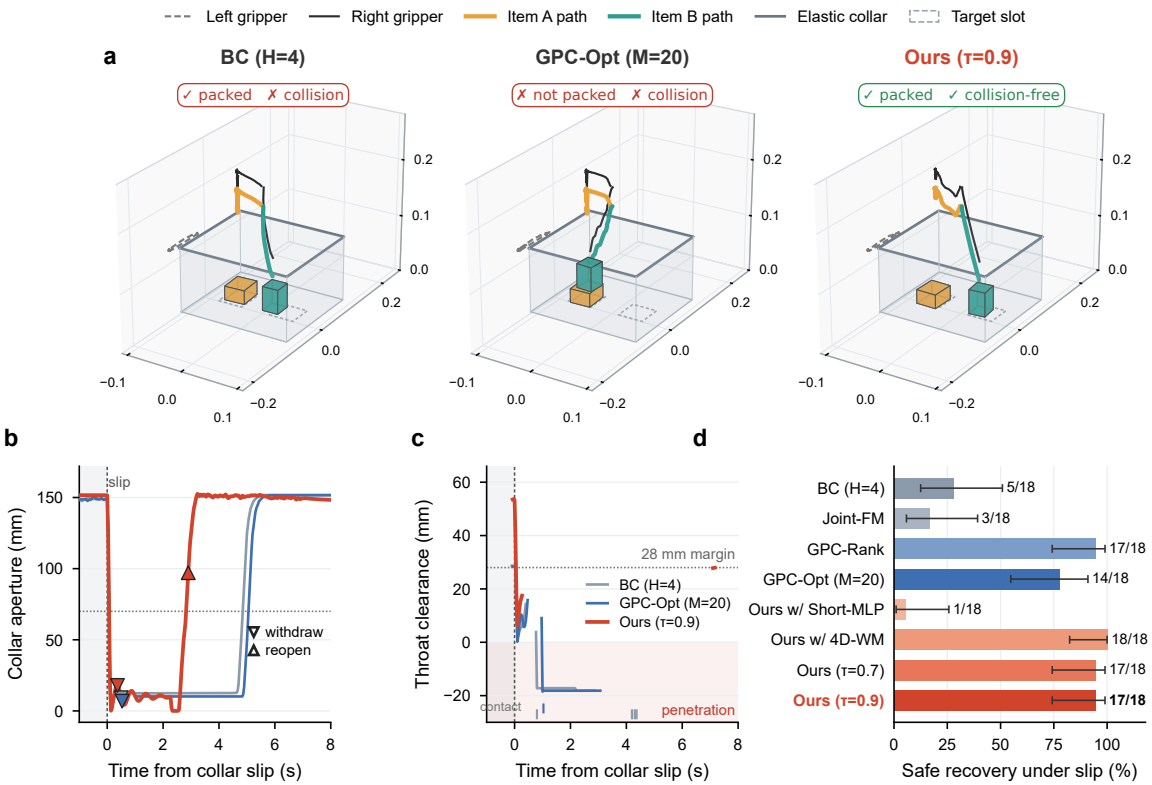

Left gripper
Right gripper
Item A path
Item B path
Elastic collar
Target slot
a
BC (H=4)
✓ packed ✗ collision
GPC-Opt (M=20)
✗ not packed ✗ collision
Ours (τ=0.9)
✓ packed ✓ collision-free
b
Collar aperture (mm)
slip
withdraw
reopen
Time from collar slip (s)
c
Throat clearance (mm)
28 mm margin
BC (H=4)
GPC-Opt (M=20)
Ours (τ=0.9)
contact
penetration
Time from collar slip (s)
d
BC (H=4) 5/18
Joint-FM 3/18
GPC-Rank 17/18
GPC-Opt (M=20) 14/18
Ours w/ Short-MLP 1/18
Ours w/ 4D-WM 18/18
Ours (τ=0.7) 17/18
Ours (τ=0.9) 17/18
Safe recovery under slip (%)